

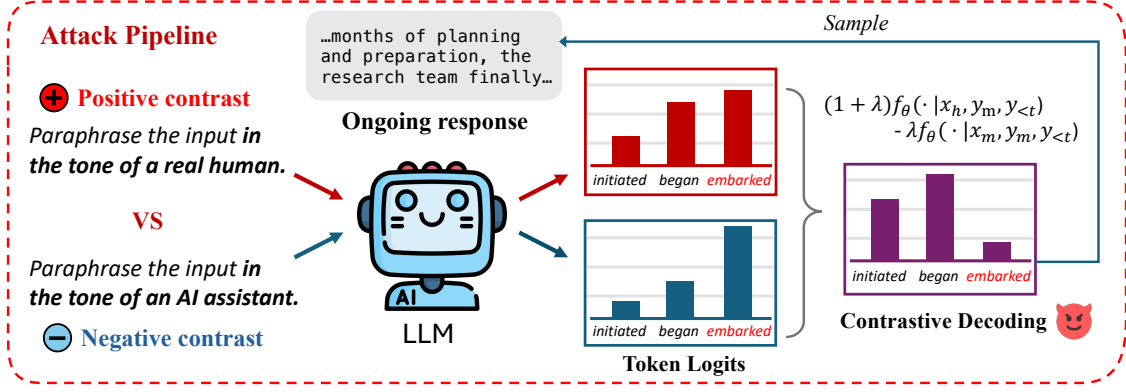


Figure 2: Overview of the proposed CoPA. The contrastive paraphrasing successfully penalizes the LLM-preferred word ‘embarked’ (Liang et al., 2024) and encourages more flexible word choices for next-token sampling.

conditional probability distribution:

$$y_t \sim p_\theta(\cdot | x, y_m, y_{<t}) \propto \exp f_\theta(\cdot | x, y_m, y_{<t}), \quad (1)$$

where $y_{<t}$ denotes the previously generated sequence. Particularly, the probability of a generated text y with length l can be expanded as the multiplication of conditional probabilities:

$$q_\theta(y) = \prod_{t=1}^l p_\theta(y_t | x, y_m, y_{<t}), \quad (2)$$

where $q_\theta(\cdot)$ denotes the text probability distribution. Based on the chain rule in Eq. (2), the sentence-level paraphrasing problem can be further reformulated as a token-level selection task, *i.e.*, design algorithms to adequately penalize the probability of machine-favored tokens and inspire more human-like word choices for generating sentences able to confuse the text detector $D_w(\cdot)$. In addition to outstanding attack performance, the revised sentences should preserve the original semantics and exhibit a high degree of coherence to ensure text quality.

3.2 Contrastive Paraphrase Attacks

Building upon the preceding analysis, an intuitive approach is to devise a prompt x_h that can elicit more diverse token distributions p'_h from LLM $f_\theta(\cdot)$ to simulate authentic human-written distribution p_h . While this strategy achieves some success, we find that the inherent statistical priors of language models persistently impose constraints on the output distributions. As a result, this leads to unstable outcomes, with some revised sentences still exhibiting sufficient machine-related patterns and remaining highly detectable (see Appendix D).

To achieve more effective and stable attacks, we carefully examine this issue and identify the following two critical considerations. (1) The current strategy essentially operates within the input space (*i.e.*, modify input prompts) to indirectly influence the output token distribution, which remains inevitably constrained by the prior knowledge encoded in the LLM. Instead, directly manipulating the output distribution is a potentially more promising alternative. (2) Generating word distributions that can fully deceive detection models is challenging; however, it is much easier to generate word distributions that are highly detectable. From a reverse-thinking perspective, those LLM-favored tokens are also considerably valuable since they encapsulate rich machine-style features and can serve as negative examples for contrastive references. Based on these insights, we propose our Contrastive Paraphrase Attack, a novel approach that directly employs a dynamic adjustment to the output word distributions during LLM decoding. Figure 2 illustrates the core pipeline of CoPA. Apart from the prompt x_h for human-style distributions p'_h , we also construct a comparative machine prompt x_m to elicit machine-preferred word choices p_m . By contrasting human-like and machine-like token distributions, CoPA refines the probabilities in the decoding process and encourages generations of texts with enhanced human-written resemblance. Formally, the contrastively purified token distribution at timestep t can be expressed as:

$$p_c(\cdot | x_h, x_m, y_m, y_{<t}) \propto \exp \left((1 + \lambda) f_\theta(\cdot | x_h, y_m, y_{<t}) - \lambda f_\theta(\cdot | x_m, y_m, y_{<t}) \right), \quad (3)$$

where p_c represents the contrastive token distributions and λ is the scaling parameter that controls the

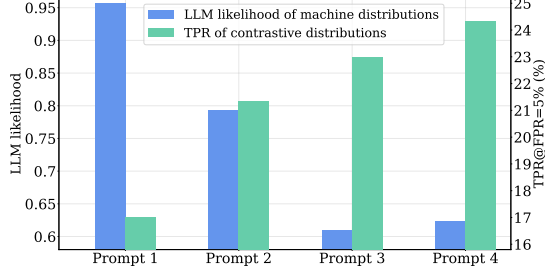


Figure 3: Fast-DetectGPT detected LLM likelihood of texts from different machine distributions p_m induced by various machine prompts x_m . We also present the detection TPR of their corresponding contrastive distributions p_c . See Appendix E for details of used prompts.

degree of amplification on the discrepancy between two distributions. The proposed framework operates as a self-corrective decoding mechanism that is specifically designed to trick AI-text detectors. By dynamically identifying and penalizing machine-preferred token preferences, CoPA effectively reduces entrenched linguistic biases and enables the generation of sentences with more expressiveness and lexical diversity, achieving an enhanced capability to fool LLM-generated text detectors.

Reduce machine styles by amplifying them.

Previous studies have shown that rewriting machine texts using another LLM can reduce some machine-specific features, although not sufficiently for launching effective attacks (Sadasivan et al., 2023). This is because the rewritten texts inherit a mixture of stylistic and lexical preferences from different LLMs, thus increasing the text diversity. However, in our contrastive design, the machine-style distribution p_m actually serves as a negative reference to be subtracted from the human distribution. Employing a regular paraphrasing prompt to obtain p_m may inadvertently dilute its machine-specific characteristics and weaken the effectiveness of the contrastive operation.

A reasonable solution involves constructing p_m as a highly salient and concentrated machine-style token distribution, wherein high-probability tokens are strongly machine-related and more likely to trigger detection. We achieve this by identifying a prompt x_m that can amplify the LLM likelihood of generated sentences. Fig. 3 empirically validates our strategy, *i.e.*, magnifying machine-style features in p_m can, in turn, promote a refined distribution that more closely resembles authentic human writing, further enhancing the attack effectiveness.

Adaptive Truncation for plausibility. Another important issue is that CoPA utilizes the whole

token distribution to measure the difference. However, there may be occasions where certain high-probability tokens overlap between p'_h and p_m . The subtraction operation may penalize the probabilities of reasonable and valid tokens while rewarding casual and unrelated ones (Fang et al., 2025), thus compromising the coherence and semantic consistency of generated sentences. To address this, we incorporate a token constraint mechanism (Li et al., 2023), which applies an adaptive pruning to the output tokens:

$$y_t \sim p_c(\cdot|x_h, x_m, y_m, y_{<t}), \text{ s.t. } y_t \in \mathcal{V}_{top}(y_{<t}),$$

$$\mathcal{V}_{top}(y_{<t}) = \left\{ y_t \in \mathcal{V} : p'_h(y_t|x_h, y_m, y_{<t}) \geq \alpha \max_v p'_h(v|x_h, y_m, y_{<t}) \right\}, \quad (4)$$

where $\mathcal{V}$ denotes the vocabulary set of f_θ and α is the hyperparameter to adjust clipping. By introducing this adaptive pruning mechanism, CoPA leverages the confidence scores of the human-like distribution to refine the contrastive distribution, which restricts the decision-making to a more reliable token candidate pool and suppresses the selection of unsuitable tokens.

3.3 Theoretical Analysis

Apart from empirical analysis, we build a theoretical framework to confirm the superiority of CoPA in simulating authentic human writing. As previously stated, p_h denotes the real human-chosen word distribution, p'_h and p_m represent human-like and machine-like token distributions elicited from the LLM using prompts x_h and x_m , respectively. The objective is to prove that the distribution p_c , derived by contrasting p'_h and p_m , aligns more closely with the human preferences p_h .

To mathematically measure the difference between distributions p_h and p_c , we first introduce an auxiliary function based on the KL divergence.

Definition 1 (Auxiliary Distance Function). *Let $\mathbb{KL}$ denotes the KL divergence, the distributional distance between p_h and p_c is a unary function of λ , which is characterized as*

$$g(\lambda) := \mathbb{KL}(p_h || (1 + \lambda)p'_h - \lambda p_m). \quad (5)$$

$g(\lambda)$ inherit several good properties from the KL divergence, based on which we derive the critical Theorem that guarantees the effectiveness of CoPA:

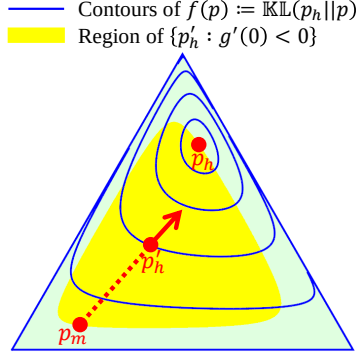


Figure 4: Illustration of the premise of Theorem 1. Let $|\mathcal{V}| = 3$ and thus $\mathbb{P}^{\mathcal{V}}$ is a triangle. We draw the contours of $f(p) := \mathbb{KL}(p_h || p)$. The closer to p_h , the lower the KL divergence with p_h . If $p'_h - p_m$ points to the inside of the contour at p'_h , then $f(p)$ decreases at p'_h along $p'_h - p_m$, i.e., $g(\lambda)$ decreases at $\lambda = 0$. In this case $g'(0) < 0$ is satisfied and Theorem 1 is applicable. In practice p'_h is usually between p_m and p_h , so $g'(0) < 0$ is usually satisfied and Theorem 1 is generally applicable.

Proposition 1. $g(\lambda)$ is a convex function. If $g(\lambda)$ is not constant, it has a unique minimum point λ_* .

Theorem 1. If $g'(0) < 0$, then $\lambda_* > 0$ and for any $\lambda \in (0, \lambda_*]$, we have

$$\mathbb{KL}(p_h || (1 + \lambda)p'_h - \lambda p_m) < \mathbb{KL}(p_h || p'_h). \quad (6)$$

The detailed proofs of Proposition 1 and Theorem 1 are provided in Appendix A. Theorem 1 reveals that by adequately selecting λ , CoPA drives the resultant distribution p_c closer to the authentic human distribution p_h than the human-like distribution p'_h , which is directly elicited from the LLM using x_h . This theoretically validates the necessity and effectiveness of our contrastive strategy.

Note that the premise of Theorem 1 is $g'(0) < 0$. We use $|\mathcal{V}| = 3$ as an example to illustrate its rationality. As in Figure 4, we calculate the area wherein probability distributions satisfy $g'(0)' < 0$. In essence, p'_h is generated by the LLM and hence is constrained by the inherent language priors, which prevent it from deviating significantly from the machine-featured distribution p_m . Meanwhile, we use a carefully crafted human-like prompt to guide p'_h to move from p_m towards p_h . As a result, p'_h typically falls within the region that satisfies $g'(0) < 0$. Therefore, this premise generally holds and thus Theorem 1 is applicable in practice, which is further confirmed by experimental results in Sec. 4.2. Based on LLM’s prediction paradigm, we contrast the output logits in practice, achieving excellent performance in misleading detection models. We

also note that researchers should examine the validity of this assumption in their specific setting before applying our theoretical framework.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on three widely adopted datasets spanning various linguistic styles and content, including (1) XSum for news articles (Narayan et al., 2018), (2) SQuAD for Wikipedia contexts (Rajpurkar, 2016), and (3) LongQA for long-form question answering, where LLM answers a how/why question within 250-350 words (Fan et al., 2019). We follow (Bao et al., 2024) and randomly select 150 samples for evaluation.

Baselines. We compare our method with the state-of-the-art (SOTA) surrogate-free paraphrase attack Dipper (Krishna et al., 2023). We adopt the setup of 60 lexical diversity and 60 order diversity for Dipper to achieve its best performance. Additionally, we reproduce the attack introduced in (Mao et al., 2024), which leverages an LLM with the query “Help me rephrase it in human style.” to rewrite machine texts (denoted as Raidar-A). For fairness, we use the same LLM for both our paraphraser and baselines. Note that we also reveal the superiority of CoPA over (Shi et al., 2024) that relies on an extra surrogate model in Section C.

For detection algorithms, we consider diverse methods including training-free LogRank (Solaiman et al., 2019), DetectGPT (Mitchell et al., 2023), DNA-GPT (Yang et al., 2024), Fast-DetectGPT (Bao et al., 2024), Raidar (Mao et al., 2024), TOCSIN (Ma and Wang, 2024), and training-based RoBERTa (Liu, 2019) provided by OpenAI and the R-detect (Song et al., 2025).

Metrics. We analyze the performance using two key metrics. (1) Detection accuracy. In real-world applications, it is crucial to guarantee that human-written text should almost never be misclassified as machine-generated (Krishna et al., 2023), i.e., satisfying a very low false positive rate (FPR). Hence, we follow Dipper and report the true positive rate (TPR) at a fixed FPR. Specifically, we set a relatively high FPR of 5% to significantly reveal the performance improvements. Please refer to Appendix C for results with FPR=1%. (2) Semantic similarity. The rewritten sentences should preserve the original semantics. Similar to Dipper, we employ the P-SP (Wieting et al., 2022) model, a specialized embedding model trained on a filtered